Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Jim Vieira and Son Dairy #2

Physical address of dairy:

Number and Street	City	County	Zip Code
3100 S Prairie Flower RD	Turlock	Stanislaus	95380

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1960

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X058-X014-X024-XXXX X058-X014-X025-XXXX

B. OPERATORS

Vieira, Jim

Operator name: Vieira, Jim

Telephone no.: (209) 668-2244 (Landline) (209) 652-6980 (Cellular)

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 2397	Turlock	CA	95381

This operator is responsible for paying permit fees.

C. OWNERS

Machado, Lucio

Legal owner name: Machado, Lucio

Telephone no.: (209) 632-4838 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
3100 N Prairie Flower RD	Turlock	CA	95380

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	0	0	310	90	0	0
Maximum number	0	0	310	90	0	0
Average number	0	0	310	90	0	0
Avg live weight (lbs)	0	0	1,000	800		

Predominant milk cow breed: Holstein

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 4,122.29 *tons per reporting period*

Total nitrogen from manure: 37,960.00 *lbs per reporting period*

After ammonia losses (30% loss applied): 26,572.00 *lbs per reporting period*

Total phosphorus from manure: 6,424.00 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 3,240,000 *gallons*

Total nitrogen generated: 13,114.33 *lbs*

Total phosphorus generated: 3,837.87 *lbs*

Total potassium generated: 41,808.45 *lbs*

Total salt generated: 175,495.35 *lbs*

	3,240,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	3,240,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
TID Lateral 5	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Lucios	29	29	2	both	X058-X014-X024-XXXX X058-X014-X025-XXXX
Totals for areas that were used for application	29	29	2		
Totals for areas that were not used for application					
Land application area totals	29	29	2		

B. CROPS AND HARVESTS

Lucios

Field name: Lucios

10/27/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 29 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	703.50 *ton*	As-is		71.4	3,500.00	700.00	5,100.00		7.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	24.26	169.81	33.96	247.44	985.19

06/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 29 Plant date: 06/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	902.77 *ton*	As-is		66.4	4,900.00	1,100.00	6,200.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	31.13	305.07	68.49	386.01	1,338.84

NUTRIENT BUDGET

A. LAND APPLICATIONS

Lucios - 10/27/2022: Oats, silage-soft dough

Field name: Lucios

Crop: Oats, silage-soft dough — Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	92.11	26.02	271.59	1,215.49	660,000.00 *gal*
TID Lateral 5	Surface water	0.00	0.00	0.00	269.21	4,455,000.00 *gal*
Application event totals		92.11	26.02	271.59	1,484.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	131.56	32.63	374.66	1,674.76	600,000.00 *gal*
TID Lateral 5	Surface water	0.00	0.00	0.00	244.74	4,050,000.00 *gal*
Application event totals		131.56	32.63	374.66	1,919.49	

Lucios - 06/13/2023: Corn, silage

Field name: Lucios

Crop: Corn, silage — Plant date: 06/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	120.96	32.29	32.29	0.00	160.00 *ton*
Application event totals		120.96	32.29	32.29	0.00	

Lucios - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	172.60	49.83	626.22	2,401.63	780,000.00 *gal*
TID Lateral 5	Surface water	0.00	0.00	0.00	318.16	5,265,000.00 *gal*
Application event totals		172.60	49.83	626.22	2,719.79	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Surface water	0.00	0.00	0.00	256.98	4,252,500.00 *gal*
Application event totals		0.00	0.00	0.00	256.98	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	27.97	11.93	84.60	379.84	600,000.00 *gal*
TID Lateral 5	Surface water	0.00	0.00	0.00	244.74	4,050,000.00 *gal*
Application event totals		27.97	11.93	84.60	624.58	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Surface water	0.00	0.00	0.00	244.74	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	244.74	

08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Surface water	0.00	0.00	0.00	256.98	4,252,500.00 *gal*
Application event totals		0.00	0.00	0.00	256.98	

Lucios - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	27.97	11.93	84.60	379.84	600,000.00 *gal*
TID Lateral 5	Surface water	0.00	0.00	0.00	244.74	4,050,000.00 *gal*
Application event totals		27.97	11.93	84.60	624.58	

09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Surface water	0.00	0.00	0.00	232.50	3,847,500.00 *gal*
Application event totals		0.00	0.00	0.00	232.50	

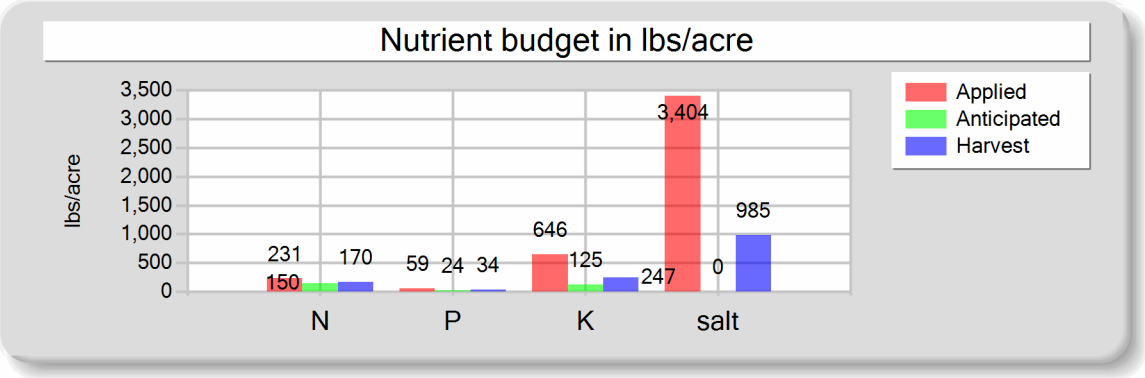
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Surface water	0.00	0.00	0.00	220.26	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	220.26	

B. NUTRIENT BUDGET

Lucios - 10/27/2022: Oats, silage-soft dough

Field name: Lucios | Crop: Oats, silage-soft dough | Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	223.67	58.65	646.25	2,890.25
Fresh water	0.00	0.00	0.00	513.95
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	230.67	58.65	646.25	3,404.20
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	169.81	33.96	247.44	985.19
Nutrient balance	60.86	24.69	398.81	2,419.01
Applied to removed ratio	1.36	1.73	2.61	3.46

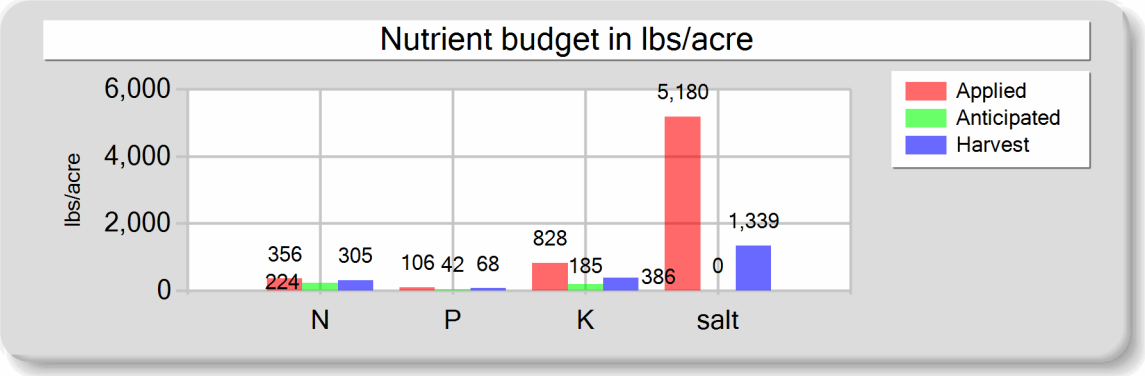
Fresh water applied	
8,505,000.00	*gallons*
313.21	*acre-inches*
10.80	*inches/acre*

Process wastewater applied	
1,260,000.00	*gallons*
46.40	*acre-inches*
1.60	*inches/acre*

Total harvests for the crop	
1	*harvests*

Lucios - 06/13/2023: Corn, silage

Field name: Lucios | Crop: Corn, silage | Plant date: 06/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	120.96	32.29	32.29	0.00
Process wastewater	228.54	73.69	795.42	3,161.32
Fresh water	0.00	0.00	0.00	2,019.09
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	356.50	105.98	827.71	5,180.41
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	305.07	68.49	386.01	1,338.84
Nutrient balance	51.43	37.49	441.70	3,841.57
Applied to removed ratio	1.17	1.55	2.14	3.87

Fresh water applied	
33,412,500.00	*gallons*
1,230.47	*acre-inches*
42.43	*inches/acre*

Process wastewater applied	
1,980,000.00	*gallons*
72.92	*acre-inches*
2.51	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 05/18/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 50.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,100.00	5,900.00	5,900.00	0.00	0.00	0.00	0.00	0.00		0.00
DL	0.01	0.01	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,100.00	13,000.00	26,200.00							0.00
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

WWQ4

Sample and source description: WWQ4

Sample date: 10/13/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	485.00	303.00	0.00	0.00	137.00	1,430.00								10,200.00	6,400
DL	1.00	0.50	0.50	0.50	0.10	0.50								10.00	10

WWQ1

Sample and source description: WWQ1

Sample date: 02/23/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	762.00	366.00	0.00	0.00	189.00	2,170.00								13,600.00	9,700
DL	0.50	0.50	0.50	0.50	0.10	0.50								10.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 05/18/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	769.00	369.00	0.00	0.00	222.00	2,790.00	378.00	105.00	743.00	5,360.00	0.00	5.70	1,660.00	15,700.00	10,700
DL	1.00	0.50	0.50	0.50	0.10	0.50	0.10	0.10	1.00	5.00	1.00	0.50	0.20	10.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 08/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.90

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	162.00	90.00	0.00	0.00	69.10	490.00								3,140.00	2,200
DL	1.00	0.50	0.50	0.50	0.10	0.50								10.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 11/09/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	293.00	212.00	0.00	0.00	80.40	784.00								5,680.00	3,850
DL	1.00	0.50	0.50	0.50	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

TID Lateral 5

TID Lateral 5

Sample description: TID Lateral 5

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	210
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Lucios - 10/27/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,500.00	700.00	5,100.00		7.10
DL	100.00	100.00	100.00		0.01

Lucios - 06/13/2023: Corn, silage

Lucios - 06/13/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,900.00	1,100.00	6,200.00		6.40
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

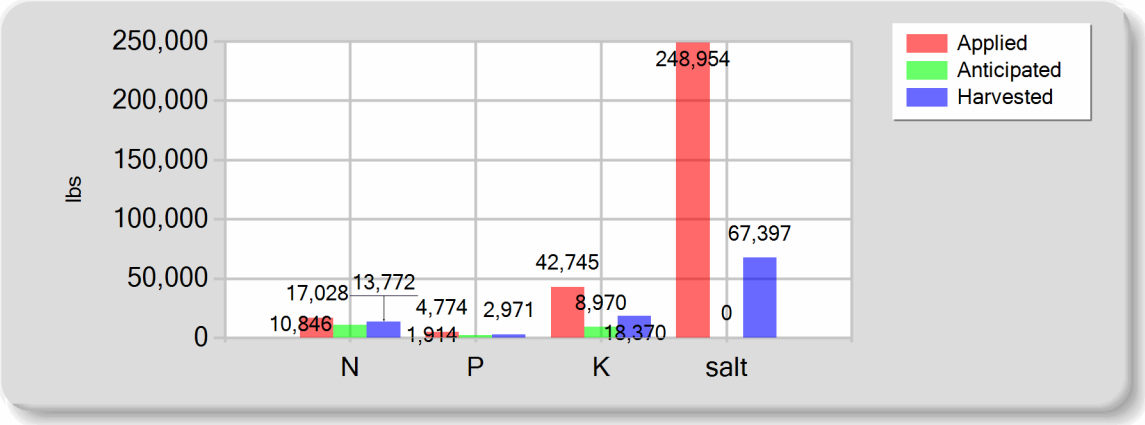
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

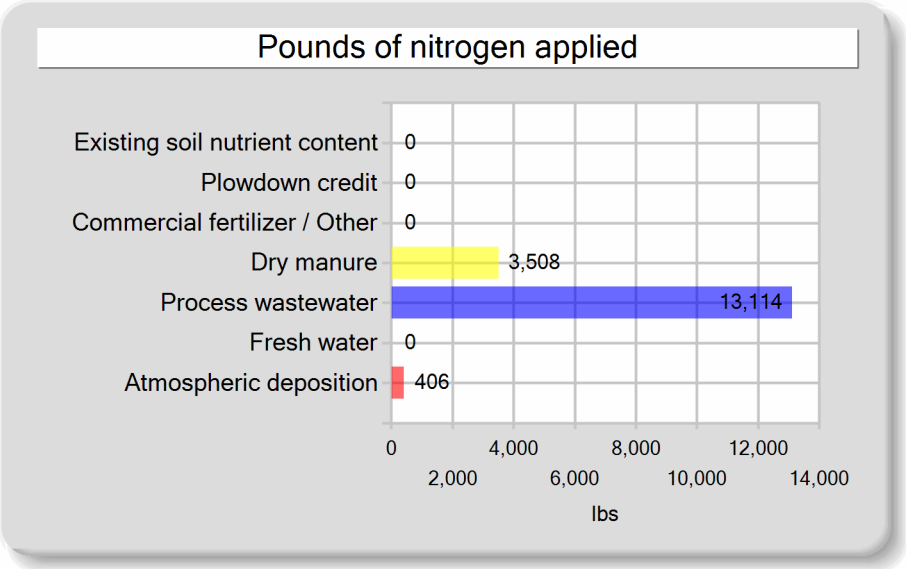
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

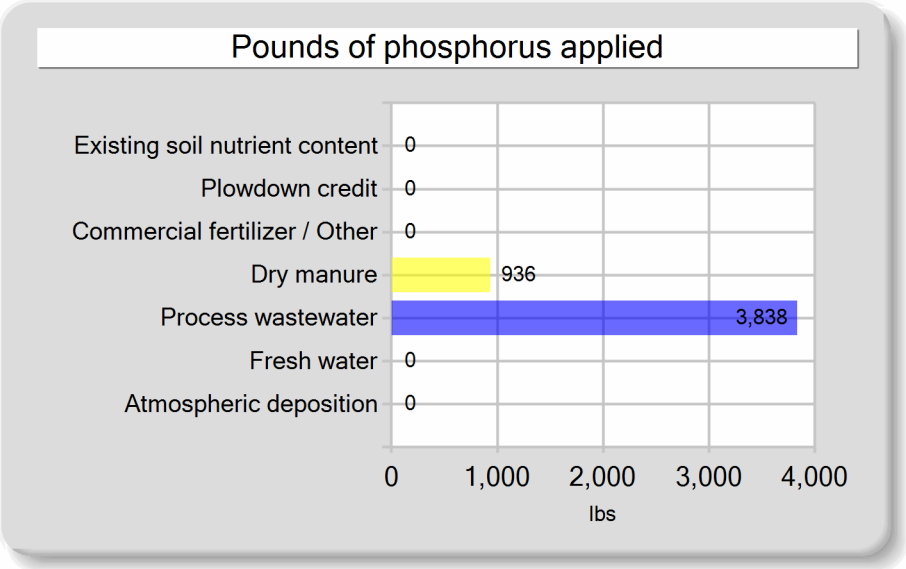
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	3,507.71	936.45	936.45	0.00
Process wastewater	13,114.33	3,837.87	41,808.45	175,495.35
Fresh water	0.00	0.00	0.00	73,458.32
Atmospheric deposition	406.00	0.00	0.00	0.00
Total nutrients applied	17,028.05	4,774.31	42,744.90	248,953.67
Anticipated crop nutrient removal	10,846.00	1,914.00	8,969.70	0.00
Actual crop nutrient removal	13,771.65	2,970.99	18,370.05	67,396.87
Nutrient balance	3,256.40	1,803.32	24,374.85	181,556.80
Applied to removed ratio	1.24	1.61	2.33	3.69

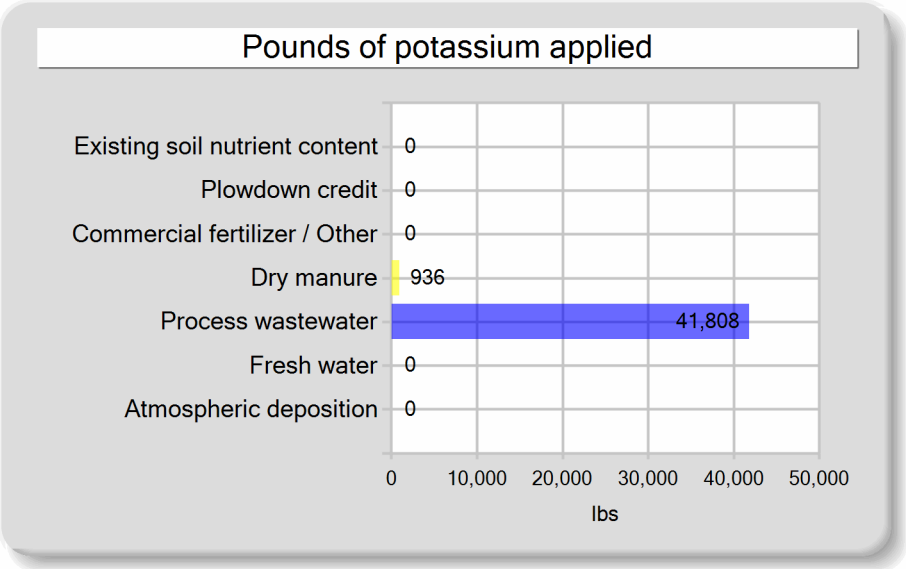
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

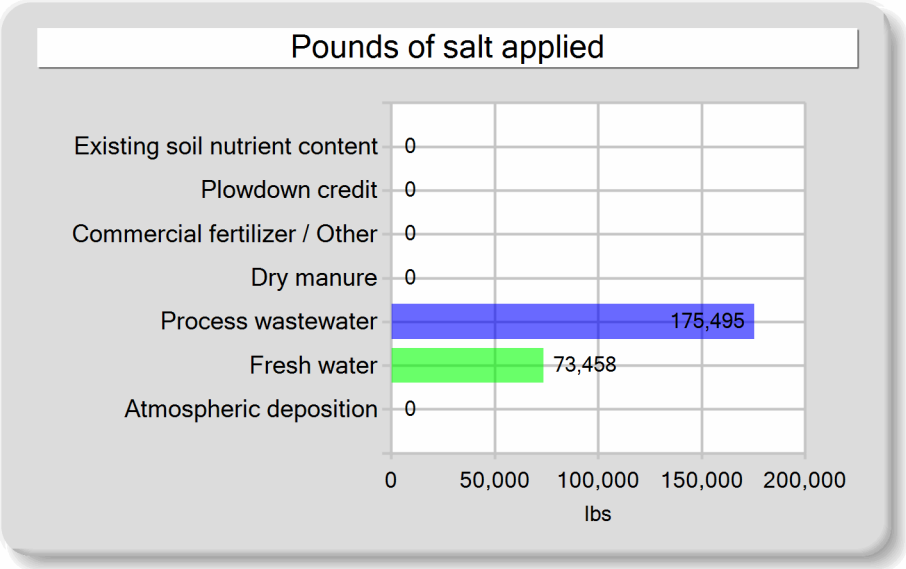


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Lucio Machado	Jim Vieira
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Lucio Machado 4-18-2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DELLAVALLE™
LABORATORY INC

GJC Dairy
P.O. Box 2397
Turlock, CA 95381

Account# 00-0025382
Account Manager: Ben Nydam
Submitted By: F & R Ag Services
Ranch: Dairy #2

Received: 12/13/2023 15:26
Reported: 12/18/2023 15:26

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L0841-01	Dairy 2	Drinking Water	Jim Vieira		12/13/2023 7:30

Default Cooler Temperature on Receipt °C: -1.2
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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1910 W. McKinley Ave Suite 110 Fresno, CA 93728 559-233-6129 www.dellavallelab.com

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LABORATORY INC

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Ranch: Dairy #2

Received: 12/13/2023 15:26
Reported: 12/18/2023 15:26

Sample Results

Sample: Dairy 2
23L0841-01 (Water)

Sampled: 12/13/2023 7:30
Sampled By: Jim Vieira

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.10**	mmhos/cm	0.01	1		12/14/23 19:36	SM 2510 B		BEL0659
Electrical Conductivity umhos	**1100**	umhos/cm	10.0	1		12/14/23 19:36	SM 2510 B		BEL0659
Ammonia (as N)	*	mg/L	0.00	1		12/13/23 07:30	Field		BEL0585
Nitrate Nitrogen as NO3N	**28.8**	mg/L	0.1	1	10	12/15/23 04:58	EPA 300.0		BEL0594
Temperature	**25.0**	units	0.0	1		12/14/23 19:36	SM 4500-H+	H	BEL0659
pH	**6.8**	units	1.0	1		12/14/23 19:36	SM 4500-H+	H	BEL0659

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Received: 12/13/2023 15:26
Reported: 12/18/2023 15:26

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0594										
Blank (BEL0594-BLK1)					Prepared & Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0594-BLK2)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0594-BLK3)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0594-BLK4)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0594-BLK5)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEL0594-BS1)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	4.6		0.1	mg/L	5.000		91.6	90-110		
LCS (BEL0594-BS2)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		105	90-110		
LCS (BEL0594-BS3)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	4.6		0.1	mg/L	5.000		92.1	90-110		
LCS (BEL0594-BS4)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	4.7		0.1	mg/L	5.000		93.9	90-110		
Duplicate (BEL0594-DUP1)		**Source: 23L0838-01**			Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	8.5		0.1	mg/L		8.4			0.603	10
Duplicate (BEL0594-DUP2)		**Source: 23L0856-02**			Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	2.0		0.1	mg/L		1.9			1.63	10
Duplicate (BEL0594-DUP3)		**Source: 23L0892-03**			Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	4.5		0.1	mg/L		4.6			1.08	10
Duplicate (BEL0594-DUP4)		**Source: 23L0910-05**			Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	8.0		0.1	mg/L		8.0			0.212	10
Matrix Spike (BEL0594-MS1)		**Source: 23L0838-01**			Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	13.8		0.1	mg/L	5.000	8.4	107	90-110		
Matrix Spike (BEL0594-MS2)		**Source: 23L0856-02**			Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	6.8		0.1	mg/L	5.000	1.9	96.3	90-110		

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Received: 12/13/2023 15:26
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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0594 (Continued)										
Matrix Spike (BEL0594-MS3)			**Source: 23L0892-03**		Prepared: 12/14/2023	Analyzed: 12/15/2023				
Nitrate Nitrogen as NO3N	9.4		0.1	mg/L	5.000	4.6	95.9	90-110		
Matrix Spike (BEL0594-MS4)			**Source: 23L0910-05**		Prepared: 12/14/2023	Analyzed: 12/15/2023				
Nitrate Nitrogen as NO3N	12.9		0.1	mg/L	5.000	8.0	97.7	90-110		
Reference (BEL0594-SRM1)					Prepared & Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	9.2			mg/L	10.00		92.2	90-110		
Reference (BEL0594-SRM2)					Prepared: 12/14/2023	Analyzed: 12/15/2023				
Nitrate Nitrogen as NO3N	9.3			mg/L	10.00		92.7	90-110		
Reference (BEL0594-SRM3)					Prepared: 12/14/2023	Analyzed: 12/15/2023				
Nitrate Nitrogen as NO3N	9.3			mg/L	10.00		93.3	90-110		
Reference (BEL0594-SRM4)					Prepared: 12/14/2023	Analyzed: 12/15/2023				
Nitrate Nitrogen as NO3N	9.4			mg/L	10.00		93.6	90-110		
Reference (BEL0594-SRM5)					Prepared: 12/14/2023	Analyzed: 12/15/2023				
Nitrate Nitrogen as NO3N	9.4			mg/L	10.00		94.1	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0659										
Blank (BEL0659-BLK1)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.5		1.0	units						
Blank (BEL0659-BLK2)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.0		1.0	units						
Blank (BEL0659-BLK3)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
pH	7.7		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEL0659-DUP1)		**Source: 23L0838-09**			Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	1.04		0.01	mmhos/cm		1.03			0.618	10
pH	7.2		1.0	units		7.2			0.695	10
Electrical Conductivity umhos	1040		10.0	umhos/cm		1030			0.618	10
Duplicate (BEL0659-DUP2)		**Source: 23L0852-04**			Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	0.89		0.01	mmhos/cm		0.90			1.22	10
Electrical Conductivity umhos	893		10.0	umhos/cm		904			1.22	10
pH	8.0		1.0	units		8.0			0.500	10
Reference (BEL0659-SRM1)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	429			umhos/cm	426.0		101	90-110		
Reference (BEL0659-SRM2)					Prepared & Analyzed: 12/14/2023					
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0659-SRM3)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	1020			umhos/cm	1000		102	90-110		
Electrical Conductivity umhos	1020			umhos/cm	1000		102	90-110		
Reference (BEL0659-SRM4)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	1050			umhos/cm	1000		105	90-110		
Electrical Conductivity umhos	1050			umhos/cm	1000		105	90-110		
Reference (BEL0659-SRM5)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		

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Ranch: Dairy #2

Received: 12/13/2023 15:26
Reported: 12/18/2023 15:26

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0659 (Continued)										
Reference (BEL0659-SRM5)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEL0659-SRM6)					Prepared & Analyzed: 12/14/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0659-SRM7)					Prepared & Analyzed: 12/14/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0659-SRM8)					Prepared & Analyzed: 12/14/2023					
pH	4.0			units	4.000		100	97.5-102.5		

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12/13/23 15:26 23L0841

JG

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

WORK REQUEST

No. of Water Samples 1 No. of Bottles 1

Water Type: [X] Drinking Water [ ] Wastewater [ ] Ag Water [ ] Groundwater [ ] Monitoring Well [ ] Supply Water

Other: No. of Samples **Soil** ____ **Tissue** ____ **Manure** ____

Logged By ____ Results Needed By ____

BILL TO: 253862 / 08 (Acct#/Cons.) ____ (Purchase Order No)

GJC DAIRY
Company Name

Address
TURLOCK CA
City, State Zip

Telephone ____ Fax ____

Cell/Email

REQUESTED BY F&R Ag Services

PROJECT DAIRY #2

Crop ____ Intended ____

Sampled By JIM VIEIRA

Notes: Jamie - 12/18/23 13:17 - no ph# on file for F&R Ag rep, sent email to Sean to call ~~the~~ this for results/12/18/23 13:33
Per Sean, OK 4 thanks

Analysis and Bottles Required:

[X] Well Water **(DWW1)** - EC, NO_3-N, Field NH_4-N, if detected run NH_4-N (1-1 liter plastic, unpreserved) **(Annually)**
[ ] Well Water **(DWW2)** - DWW1 plus HCO_3, CO_3, Cl, SO_4, Dissolved: Ca, Mg, Na, TDS (1-1 liter plastic, unpreserved) **(Every 5 yrs)**
[ ] Process Water **(DPW1)** - EC, NO_3-N, NH_4-N, TKN, TDS, TP, TK (1-1 liter plastic, unpreserved) **(Quarterly)**
[ ] Process Water **(DPW2)** - DPW1 plus CO_3, HCO_3, Cl, SO_4, Total: Ca, Mg, Na, (1-1 liter plastic, unpreserved) **(Every 2 yrs)**
[ ] Canal Water **(DCW1)**-EC, NO_3-N,TKN,TN,TDS (1-1 liter plastic,unpreserved)
[ ] Water – EC, NO_3-N, TDS, TP (1-1 liter plastic, unpreserved)
[ ] Manure **(DM1)** - N, P, K, Moisture **(2x/yr)**
[ ] Manure **(DM2)** - DM1 plus Ash, Ca, Mg, Na, S, Cl **(Every 2 yrs)**
[ ] Compost **(OSA2)** – Moisture, N, P, K, Ash, OM, C:N ratio, Ca, Mg, Na, Fe, Cu, Mn, Zn, S
[ ] Harvest Tissue **(CRA1)** - Moisture, N, P, K, Ash
[ ] Tissue – % Moisture
[ ] Fall Soil – **(DS2)** - FA1 (0-1') **plus CEC, OM**; NO_3-N (1-2')
[ ] OTHER:

COPIES TO: labs@fragservices.com, slocke@fragservices.com
[ ] Co Health Dept
[ ] RWQCB [X] Copy of Chain
[ ] State Forms [ ] QA/QC Documents

Description of Sample	Field NH_4-N	Date/Time Sampled	Rec'd Temp °C	Description of Sample	Field NH_4-N	Date/Time Sampled	Rec'd Temp °C
1. DAIRY 2		12/13 7:30	~~3.6~~ 1.2 JG	6.			
2.							
3.				8.			
4.				9.			
5				10.			

28.9

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First		F+R	12/13 11:00	
Second	JLP	DLV	12-13-23 1:21	12-13-23 3:01
Third				
Fourth		DLI	12/13/23 15:26	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days, overdue accounts will be charged a dated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Shipping
$ ____ In
$ ____ Out

Amt Paid ____ Rec By ____ Check # ____ Date ____

Signature ____

NO PERSONAL CHECKS ACCEPTED

Paste Label Here

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

☐ Samples refridgerated before pick up | ☐ Picked up samples placed in Ice chest

Container: Ice Chest ☒ Box ☐ None ☐ | **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

	Type of Container(s) Received	Sample Number 1	2	3	4	5	6	7	8	9	10
	Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
Plastics	250 mL unpreserved (White) Plastic										
Plastics	250 mL HNO_3 (Red) Plastic										
Plastics	* pH Value										
Plastics	250 mL H_2SO_4 (Yellow) Plastic										
Plastics	* pH Value										
Plastics	500 mL unpreserved (White) Plastic										
Plastics	1 L unpreserved (White) Plastic										
Plastics	1 L unpreserved (BOD) (Purple) Plastic										
Special	500mL unpreserved (White) Glass										
Special	PO4-P Kit										
Special	**Other:**										
	Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
Plastics	250 mL unpreserved (White) Plastic										
Plastics	250 mL HNO_3 (Red) Plastic										
Plastics	250 mL H_2SO_4 (Yellow) Plastic										
Plastics	500 mL HNO_3 (Red)										
Plastics	1 L unpreserved (White) Plastic										
Plastics	1 L unpreserved (BOD) (Purple) Plastic										
Plastics	1 L HNO_3 (Red)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ + MCAA (EPA531)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ (EPA547)										
VOA Vials	40mL AG VOA unpreserved (White) (Set of 3)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
VOA Vials	40mL VOA, H_3PO_4 (Set of 3)										
VOA Vials	40 mL VOA, HCl (Blue) (Set of 3)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
Glass	250 mL AG H_2SO_4 (Yellow)										
Glass	250 mL AG $Na_2S_2O_3$ (Green)										
Glass	250 mL AG $Na_2S_2O_3$ + MCAA										
Glass	500 mL glass unpreserved (White)										
Glass	500 mL AG HCl (Blue)										
Glass	1 L AG unpreserved (White)										
Glass	1 L AG H_2SO_4 (Yellow)										
Glass	1 L AG $Na_2S_2O_3$ (Green)										
Glass	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
Special	Cyanide - 500 mL NaOH										
Special	Asbestos - 1L P wrapped in foil (Set of 2)										
Special	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
Special	Chlorite/Bromate - 250 mL AG with EDA										
Special	HAA5 - 250mL AG Ammonium Chlorite										
Special	DO KIT										
Special	**Other:**										
Special	**Other:**										

12/13/23 15:26 23L0841

SEND OUT